

have a screen and a set-top box for that.

But HD-ready products are still expensive. What is being done to reduce their price so that they are within the reach of common consumers?

Two things will help in reducing the price. One is improving the technology. The other is market size—the more we sell, the more likely we are to improve the product-price combination.

How can broadcasters and content producers in India become ready for transition to HD?

It's going to be a very big jump for them. They are still shooting in 4:3, and HD is 16:9. So one of the transitions could be to invest in cameras and equipment for 16:9 production. Then may be the SD content could be upscaled to HD. Then they should go to HD production directly.

What makes an HDTV a 3-D-ready TV? Can any 240Hz HDTV be converted into a 3-D HDTV?

Sony has adopted the world's first 240Hz (200Hz) technology to 3-D-compatible 'BRAVIA,' combining the frame sequential method with active shutter glasses. We are reproducing 3-D images based on Sony's proprietary Motionflow 240Hz (200Hz) technology.

Since you need specific signal processing to reproduce 3-D images and your TV needs to emit signals to the glasses for them to work in sync, you cannot watch 3-D images on current 240Hz (200Hz) models that do not have the required signal processing capability.

What are the requirements for 3-D TV broadcast and reception?

At least HD-quality delivery system is required to broadcast 3-D, but even using HD, the left- and right-channel quality has to be reduced to fit the available bandwidth by using top/bottom or side-by-side formatting. On the receiver side, the left and right signals need to be reconstituted and then processed according to the display method of 3-D—it's frame sequential on Sony's BRAVIA TV.

You need an external set-top box or 3-D TV internal unit to receive and decode 3-D broadcast signals.

What are the requirements for optimal 3-D system performance?

Connection from a 3-D source player to a 3-D-enabled TV with high-speed HDMI cable.

What are the technical hurdles in broadcasting 3-D TV programming?

More 3-D content availability in the market and consumer hesitance to upgrade will act as some of the challenges to the growth of this market globally.

Without HD, there will be no 3-D. As audience will

eventually start demanding 3-D, with existing SD systems, broadcasters will find it challenging to leap to 3-D and eventually be left behind their competitors who are ahead of the curve. 3-D is going to be a gradual growth story for India but mass high-definition adoption with content will come first.

What channels are currently being broadcast in 3-D? What to expect?

3-D programming from broadcasters will be essential in ensuring the enjoyment of a wide variety of content. As such, Sony is working with leading broadcasters including ESPN and DirecTV in the US and BSkyB in the UK to ensure that audiences are capable of receiving the 3-D programming these broadcasters are currently developing.

In addition, Discovery Communications, Sony Corporation and IMAX Corporation have announced a joint venture to develop the first dedicated 3-D television network in the US. The partnership brings together three leaders with an extraordinary collection of award-winning 3-D content, technology expertise, television distribution and operational strength to deliver a high-quality 3-D viewing experience to home television audiences.

Are 3-D HDTVs compatible with standard-definition and non-3-D HD viewing? Will they be able to up-convert 2-D movies into 3-D?

Sony's newly developed proprietary technology enables 3-D-compatible BRAVIA TV sets to convert 2-D image into simulated 3-D image. If you press the 3-D button on the remote control when you're watching 2-D content, the TV detects and displays the focal points of the image, etc to the front. So viewers can feel the depth with the glasses on. However, the effect and the way viewers feel the depth vary depending on the content and the viewer.

What all issues need to be overcome for 3-D HDTV to become successful?

Technologies and techniques of making 3-D have improved significantly over the years. It is in our interest to help the industry as a whole seize this opportunity of providing consumers the real 3-D experience. For 3-D to gain acceptance in the long term, broadcasters need to consistently provide quality 3-D content that people enjoy watching.

Finally, how do you see the future of 3-D HDTVs, considering that viewers will probably don the glasses mainly for special events like sports and movies, and not necessarily to watch the evening news?

Sony has received favourable reception for 3-D products in each market. In Japan, we've maintained the top share in 3-D TV. Sony will accelerate initiatives to deliver 3-D TV for the home. Worldwide, we estimate 5 per cent of all LCD TVs to be 3-D in FY2010 and 20 per cent of all LCD TVs will be 3-D in FY 2011. Sony aims to exceed that figure. ●